# BISWANATH RANA


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## ABOUT ME

Experienced Assistant Production Manager with 3 Years of expertise in the Solar Pv cell & Module manufacturing, skilled in team leadership, problem solving, production planning and data analysis. Proficient in manpower management, achieving production targets, and cost reduction. Deep knowledge of automation tools, coupled with expertise in solar cell & Module manufacturing processes. Seeking a Managerial position in a prestigious company to leverage skills and experience towards achieving production excellence.

## EDUCATION

- **2023**

**DREAM INSTITUTE OF TECHNOLOGY (MAKAUT UNIVERSITY)**

- BACHELOR OF TECHNOLOGY IN ELECTRICAL ENGINEERING
- CGPA - 8.44

- **2020**

**SIDHU KANHU BIRSA GOVERNMENT POLYTECHNIC COLLEGE**

- DIPLOMA IN ELECTRICAL ENGINEERING
- CGPA - 7.5

## EXPERIENCE

- **Saatvik Green Energy Pvt Ltd, Ambala**
- **February 2024 - Present**

**Assistant Manager** **April 2025 - Present**

- Managed production operations in a 1.2 GW solar module manufacturing line to meet daily production targets.
- Prepared and executed production plans based on customer demand and company goals.
- Monitored production data such as module breakage, downgrading, FTQ, and yield.
- Identified process problems and implemented improvements to increase efficiency and productivity.
- Worked with the maintenance team to improve machine uptime and reduce downtime.
- Identified bottlenecks in processes such as stringing, lamination, framing, and testing, and solved them.
- Conducted daily production meetings to review targets, quality issues, and line performance.
- Led and trained Shift Engineers, GETs, and Operators for better production & Quality performance.
- Coordinated with quality, procurement, logistics, and maintenance teams to ensure smooth production.
- Implemented corrective and preventive actions (CAPA) to avoid repeated defects.
- Ensured proper control of cycle time, safety standards, and shop floor discipline.
- Monitored production cost and optimized material and manpower usage.
- Supported new solar module product development and process improvement projects.
- Prepared reports on production output, quality performance, downtime, and safety.

**Senior Production Engineer** **Shift Incharge** **February 2024 - March 2025**

- Achieving production targets on a monthly, weekly, and daily basis.
- Ensured optimal use of manpower to enhance productivity.
- Analyzed machine breakdowns and implemented corrective actions to minimize downtime.
- Focused on cost reduction across manpower, machinery, and materials.
- Maintain 5S on shop floor.
- Basic Knowladge about PO closing.
- Responsible for issuing of Raw material & BOM consumption in the shop floor.

- **Premier Energies, Hydrabad** **April 2023 - February 2024**

**GET**

- Worked on semiconductor-based solar cell technologies, including PERC and TOPCon cell manufacturing processes.
- Managed and controlled wet chemistry processes such as Texturing, PSG removal, and Polishing.
- Maintained key process parameters like etch depth, reflectance, temperature, and chemical concentration to ensure stable production.
- Performed chemical bath preparation and chemical bath change procedures to maintain consistent process performance.
- Applied Statistical Process Control (SPC) tools to monitor process variation and maintain process stability.
- Analyzed production data to identify defects, wafer breakage, and yield loss, and implemented corrective actions.
- Reduced cell rejection, Cell breakage, and process defects through continuous process improvement.
- Implemented 5S, Kaizen, and continuous improvement initiatives to improve productivity and shop floor efficiency.
- Worked closely with maintenance, and quality teams to solve technical issues and improve yield.
- Prepared process performance reports and monitored KPIs such as yield, breakage rate, and process efficiency.

## PROJECT

**Saatvik Green Energy**

**Laminator Broken Modules Reduction** **Duration - Jan 2025 to Mar 2025**

Led a team at Saatvik Solar to reduce laminator broken modules, driving innovation, cost savings, and skill development.

- Reduced broken modules from 0.09% to 0.042%
- Saved approximately ₹17.28 lakhs within three months (Jan, Feb & Mar 2025).

**Premier Energy**

**Process Improvement Project – PSG Wet Chemistry Process** **Duration - Oct 2023 to Dec 2023**

Analyzed frequent chemical bath change issues in PSG wet chemistry tools during solar cell manufacturing. Identified high KOH salt formation caused by improper exhaust pressure in two PSG machines. Optimized exhaust pressure and improved process control, which increased alkaline bath life from ~50K to over 4 lakh Cells. This improvement reduced EL cell percentage, chemical consumption, and process defects, resulting in significant cost savings and higher process efficiency.

## EXPERTISE

- Leadership
- Cost Reduction
- Problem Solving
- Breakdown minimize
- Solar cell & Module manufacturing
- Project Management

## SKILLS SUMMARY

- Quality Control Cercle
- Kaizen
- Root Cause Analysis
- Safety
- 5S Management
- MS Office

## LANGUAGES

- Bengali
- Hindi
- English

## PERSONAL DETAILS

- Father's Name : Sunil Rana
- Date Of Birth : 15th September 1998
- Gender : Male.
- Marital Status : Single.
- Nationality : Indian.
- Religion : Hindu.

## DECLARATION

- I hereby declare that the information provided in my resume is true, accurate, and complete to the best of my knowledge.

**Signature - Biswanath Rana** **Date - 12/03/2026**